

Data Analysis Summary

Used libraries:

- pandas – data manipulation
- matplotlib.pyplot – generating plots
- seaborn – plots features
- from sklearn.preprocessing import MinMaxScaler – normalizing values

Data cleaning:

We observe that “TotalCharges” column has a dtype of str instead of float64. I fixed that with `df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')`

Then we check for missing values with `missing_values = df.isnull().sum()`

We find 11 missing values in “TotalCharges”. Because the tenure is 0 on all of them it means that the client didn’t make a first payment yet so the value for “TotalCharges” should be 0. I do that with `df['TotalCharges'] = df['TotalCharges'].fillna(0)`

Next we check for duplicates. We find 0 true duplicates and 22 coincidental duplicates which we don’t remove because they represent different clients.

Exploratory Analysis:

We make visual plots to better understand the data (distribution plots, correlation heatmap, boxplots for outliers, feature relationships).

Feature Engineering:

Made 3 new features:

- OptionalServices – number of optional services a client has (0 – 6)
- HighFriction – if a client has Fiber Optic but no Tech Support (0 – 1)
- LoyalMonthlyPayers – if a 2-year client still has month-to-month contract (0 – 1)

Encoded all columns with binary or one-hot encoding.

Normalized numeric features with MinMaxScaler.

Business Insights:

1. Implement a 90-day onboarding program

The tenure distribution plot revealed that there is a massive spike in churn between months 0 and 5. Clients are experiencing "buyer's remorse" or technical frustration immediately after signing up. We must deploy an automatic check-in system (calls or targeted emails) during the first 3 months to resolve technical issues before client decides to cancel.

2. Change Fiber Optic infrastructure and pricing

The churn rate by Internet Service type plot reveals that Fiber Optic clients churn at a way higher rate than DSL clients. Fiber should be a better service than DSL. The fact that the churn rate is so high means that the current price does not match the quality of the product or maybe it is overpriced.

3. Make automatic payments more comfortable, appealing.

The churn rate by method plot reveals that clients paying via "Electronic Check" churn at a very high volume compared to those using automatic alternatives. Paying manually every month forces clients to actively ask themselves if the service is worth it. Automatic payments remove this friction. Make the automatic options stand out more or offer small discounts to those who switch from Electronic Check to automatic pay.

4. Introduce "Mid-tier" pricing interventions.

Both our histogram and boxplot show us that monthly charges over \$70 lead to high churn rates. When a high-paying customer calls to cancel, offer the a customized "mid-tier" bundle (~\$60) rather than losing them entirely.

5. Target Month-to-month contract conversions at the 6-month mark.

The boxplot confirmed that month-to-month contracts are unstable. However the tenure chart showed that if a client survives past the 6 month mark the likelihood of churning drops significantly. Send an exclusive, targeted promotion to month-to-month customers exactly on their 6 month, offering them better prices if they sign a 1-year contract.

DATA LEAKAGE:

There is a problem with the "tenure" and "TotalCharges" columns. The ML model will flag every new customer as a big churn risk because they didn't have the time to increase the TotalCharges number. To fix this we drop the TotalCharges column before training a model.